

SOME SOLUTIONS TO THE PLANETARY GEOSTROPHIC EQUATIONS

TWNH

Peterson and Callies (in review) (see also *Peterson and Callies* 2023) consider how bottom-intensified vertical mixing induces a basin-scale circulation in various simple non-flat-bottomed basins. They write equations for a bottom Ekman layer and couple them to an equation for the geostrophic contour vorticity balance, in the parlance of *Waldman and Giordani* (2023), which is a type of barotropic vorticity equation.

1. GOVERNING EQUATIONS

The flow satisfies the (non-dimensional) planetary-geostrophic and hydrostatic equations:

$$\begin{aligned} (1) \quad & -fv = -\frac{\partial p}{\partial x} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right), \\ (2) \quad & fu = -\frac{\partial p}{\partial y} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right), \\ (3) \quad & \frac{\partial p}{\partial z} = b \end{aligned}$$

(see Eqs. (9)–(11) of *Peterson and Callies* in review and Eq. (3.26) of *Klinger and Haine* 2019). The flow is non-divergent:

$$(4) \quad \nabla \cdot \mathbf{u} = 0,$$

and the buoyancy field satisfies the advection-diffusion equation

$$(5) \quad \mu\rho \left(\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b \right) = \epsilon^2 \frac{\partial}{\partial z} \left(\kappa \frac{\partial b}{\partial z} \right).$$

(see Eqs. (11)–(13) of *Peterson and Callies* in review). The boundary conditions on buoyancy are:

$$(6) \quad \text{Surface } z = 0 : b = 0$$

$$(7) \quad \text{Bottom } z = -H : \frac{\partial b}{\partial z} = 0,$$

and the initial condition is $b = z$.

The diffusivity and viscosity are given by

$$(8) \quad \kappa(z) = \kappa_0 + \exp(-\alpha(z + H)),$$

$$(9) \quad \nu = \text{constant}$$

Peterson and Callies (in review) Eq. (17).

2. VERTICALLY-INTEGRATED FLOW

The vertically-integrated flow (U, V) is given by:

$$(10) \quad \begin{aligned} U &\equiv \int_{-H}^0 u \, dz = - \int_{-H}^0 z \partial_z u \, dz, \\ V &\equiv \int_{-H}^0 v \, dz = - \int_{-H}^0 z \partial_z v \, dz, \end{aligned}$$

exploiting the bottom boundary condition on (u, v) . The equations for (U, V) come from vertically integrating the momentum equations (1), (2):

$$(11) \quad -fV = - \int_{-H}^0 \frac{\partial p}{\partial x} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz$$

$$(12) \quad fU = - \int_{-H}^0 \frac{\partial p}{\partial y} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right) \, dz.$$

The depth-integrated pressure gradient in the x direction is:

$$(13) \quad \int_{-H}^0 \frac{\partial p}{\partial x} \, dz = - \frac{\partial H}{\partial x} p(x, z = -H) + \frac{\partial}{\partial x} \int_{-H}^0 p \, dz,$$

$$(14) \quad = -p_b \frac{\partial H}{\partial x} + \frac{\partial}{\partial x} \left[p_b H - \int_{-H}^0 z \frac{\partial p}{\partial z} \right],$$

$$(15) \quad = H \frac{\partial p_b}{\partial x} - \frac{\partial}{\partial x} \int_{-H}^0 z b \, dz,$$

using the Liebniz integration formula¹, where $p_b = p(x, -H(x))$ is the bottom pressure, and integration by parts². The viscous term in the x direction is:

$$(16) \quad \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz = \left[\nu \frac{\partial u}{\partial z} \right]_{-H}^0 = \frac{\tau^x}{\epsilon^2} - \nu \frac{\partial u}{\partial z} \Big|_{z=-H},$$

which uses the boundary condition (31).

Therefore (11) and (12) lead to:

$$(17) \quad -fV = -H \frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x,$$

$$(18) \quad fU = -H \frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y,$$

¹The Liebniz integration formula reads:

$$\partial_x \left(\int_{a(x)}^{b(x)} f(x, y) \, dy \right) = \partial_x b f(x, b(x)) - \partial_x a f(x, a(x)) + \int_{a(x)}^{b(x)} \partial_x f(x, y) \, dy.$$

²Integration by parts gives:

$$\begin{aligned} \int_{-H}^0 z \frac{\partial p}{\partial z} \, dz &= [zp]_{-H}^0 - \int_{-H}^0 p \, dz, \\ &= p_b H - \int_{-H}^0 p \, dz. \end{aligned}$$

where

$$(19) \quad \chi = - \int_{-H}^0 z b \, dz$$

is the baroclinic potential energy, and $\epsilon^2 \nu \partial_z(u, v)|_{z=-H} \equiv (\tau_b^x, \tau_b^y) = \tau_b$ is the bottom stress on the fluid.

Next, divide by water depth H :

$$(20) \quad -\frac{fV}{H} = -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H}$$

$$(21) \quad \frac{fU}{H} = -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H},$$

and take the curl³

$$(23) \quad \begin{aligned} \frac{\partial}{\partial y} \left\{ -\frac{fV}{H} \right\} - \frac{\partial}{\partial x} \left\{ \frac{fU}{H} \right\} &= \frac{\partial}{\partial y} \left\{ -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H} \right\} \\ &\quad - \frac{\partial}{\partial x} \left\{ -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H} \right\}, \end{aligned}$$

$$(24) \quad \Rightarrow \frac{\partial}{\partial y} \left\{ -\frac{f}{H} \frac{\partial \Psi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{f}{H} \frac{\partial \Psi}{\partial y} \right\} = -\frac{\partial}{\partial y} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial y} \right\} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

$$(25) \quad \Rightarrow -\frac{\partial}{\partial y} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial y} = -\frac{\partial}{\partial y} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial y} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

$$(25) \quad \Rightarrow J \left(\frac{f}{H}, \Psi \right) = J \left(\frac{1}{H}, \chi \right) - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

for streamfunction of the depth-integrated horizontal flow Ψ where

$$(26) \quad \begin{aligned} V &= \frac{\partial \Psi}{\partial x}, \\ U &= -\frac{\partial \Psi}{\partial y}, \end{aligned}$$

and Jacobian $J(A, B) = \partial_x A \partial_y B - \partial_y A \partial_x B$. Notice how this recipe (vertically integrate, divide by H , then take the curl) eliminates the bottom pressure. The final result (25) is the same as Eq. (20) of *Peterson and Callies* (in review). It is called (a simplified form of) the **geostrophic contour vorticity balance** (GCVB) by *Waldman and Giordani* (2023), although *Peterson and Callies* (in review) call it the “barotropic vorticity equation”. It is not the same equation as used by *Nøst and Isachsen* (2003) (which *Waldman and Giordani* (2023) call the “transport divergence vorticity balance”).

Notice how the GCVB (25) is an equation for the streamfunction of the depth-integrated flow Ψ given the buoyancy field (and hence the JEBAR term) and the surface and bottom stresses. The GCVB shows how depth-integrated flow Ψ across f/H contours is driven by (i) the joint effect

³Recall that

$$(22) \quad \hat{\mathbf{z}} \cdot (\nabla \times \mathbf{U}) = \partial_x V - \partial_y U.$$

of baroclinicity and relief (JEBAR), (ii) curl of wind stress divided by H , (iii) curl of the bottom stress divided by H .

3. SOLUTION TO FRICTIONAL GEOSTROPHIC EQUATIONS IN STEADY STATE

Start from (1)–(2) and write the pressure as

$$(27) \quad p(x, y, z) = p_s(x, y) + \int_z^0 b(x, y, z') dz',$$

$$(28) \quad \implies \partial_x p(x, y, z) = \partial_x p_s(x, y) + \int_z^0 \partial_x b(x, y, z') dz'$$

$$(29) \quad \implies p_b(x, y) \equiv p(x, y, z = -H(x, y)) = p_s(x, y) + \int_{-H(x, y)}^0 b(x, y, z') dz',$$

where $p_s(x, y)$ is the surface pressure (and assuming that $p_s \ll p(-H)$). Solve these frictional geostrophic equations using a Green's function, defined so that $\mathbf{u} = u + iv$, thus

$$(30) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial \mathbf{u}}{\partial z} \right) - i f \mathbf{u} = \mathbf{f},$$

$$(31) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z \mathbf{u} = \tau^x + i \tau^y,$$

$$(32) \quad \text{Bottom } z = -H : \mathbf{u} = 0,$$

where $\mathbf{f} = \frac{\partial p}{\partial x} + i \frac{\partial p}{\partial y}$ is the pressure gradient from (28).

The corresponding Green's function satisfies:

$$(33) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial G_{\mathbf{u}}}{\partial z} \right) - i f G_{\mathbf{u}} = \delta(z - \xi),$$

$$(34) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z G_{\mathbf{u}} = 0,$$

$$(35) \quad \text{Bottom } z = -H : G_{\mathbf{u}} = 0.$$

We find that the solution for constant viscosity $\nu = \nu_0$ is:

$$(36) \quad G_{\mathbf{u}}(z; \xi) = \frac{i^{3/2} e^{-\sqrt{i}\varphi(z+\xi)}}{2\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \begin{cases} -e^{2\sqrt{i}\varphi \xi} + e^{2\sqrt{i}\varphi(z+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \leq \xi, \\ -e^{2\sqrt{i}\varphi z} + e^{2\sqrt{i}\varphi(\xi+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \geq \xi, \end{cases}$$

$$\implies G_{\mathbf{u}}(z; 0) = \frac{i^{3/2} e^{-\sqrt{i}\varphi z}}{\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left[e^{2\sqrt{i}\varphi(z+H(x))} - 1 \right],$$

$$(37) \quad \implies \partial_z G_{\mathbf{u}}(z; \xi)|_{z=-H(x)} = \frac{-e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left(1 + e^{2\sqrt{i}\varphi \xi} \right),$$

$$(38) \quad \implies \partial_z G_{\mathbf{u}}(z; 0)|_{z=-H(x)} = \frac{-2e^{\sqrt{i}\varphi H(x)}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)},$$

where $\varphi = \sqrt{(f/\nu_0)}/\epsilon$ (see `Example_barotropic_GeostrophicFlow_v0.2.ipynb`). And then

$$(39) \quad \mathbf{u}(z; x) = \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi - G_u(z; 0) (\tau^x + i\tau^y),$$

The baroclinic pressure (due to b) adds to the barotropic pressure (due to $p_s(x, y)$) in the integrand of this expression. Otherwise, everything else is known (remember that x in these expressions stands for (x, y) in general).

Also, the bottom current gradient is given by

$$(40) \quad \begin{aligned} -\partial_z \mathbf{u}(z; x)|_{z=-H(x)} &= \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0\epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} \left(1 + e^{2\sqrt{i}\varphi\xi}\right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \\ &\quad - \frac{2e^{\sqrt{i}\varphi H(x)}}{\nu_0\epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} (\tau^x + i\tau^y), \end{aligned}$$

using (37) and (38), and therefore the bottom stress τ_b on the fluid is

$$(41) \quad \begin{aligned} \tau_b = -\nu_0\epsilon^2 \partial_z \mathbf{u}(z; x)|_{z=-H(x)} &= \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi}\right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \\ &\quad - \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} (\tau^x + i\tau^y). \end{aligned}$$

Notice how the derivatives of the surface pressure part of p end up multiplying expressions that are functions of x .

4. FRICTIONAL GEOSTROPHIC EQUATIONS CLOSURE WITH THE GCVB EQUATION

The problem is closed using the GCVB equation, as follows (*Salmon* 1998, p. 170). This determines the unknown surface pressure field given the surface stress (and vice versa). In particular, work from (17)–(18),

$$(42) \quad \mathfrak{U}(z; x) \equiv U + iV = \int_{-H(x)}^0 \mathbf{u}(z; x) dz,$$

$$(43) \quad = \int_{-H(x)}^0 \left\{ \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi - G_u(z; 0) (\tau^x + i\tau^y) \right\} dz.$$

Computing the z integral of the Green's function $G_u(z; \xi)^4$ gives

$$(44) \quad \begin{aligned} \int_{-H(x)}^0 G_u(z; \xi) dz &= \frac{-ie^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)}\right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1\right)}{\nu_0\epsilon^2\varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)}, \\ \Rightarrow \int_{-H(x)}^0 G_u(z; 0) dz &= \frac{-i \left(e^{\sqrt{i}\varphi H(x)} - 1\right)^2}{\nu_0\epsilon^2\varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)}. \end{aligned}$$

⁴This presumes (correctly) that the denominator in (36) does not vanish.

Therefore,

$$\begin{aligned}
 \mathfrak{U}(x) = & \frac{-i}{\nu_0 \epsilon^2 \varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} \times \\
 & \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi \xi} \left(e^{\sqrt{i}\varphi \xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \right. \\
 (45) \quad & \left. - \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 (\tau^x + i\tau^y) \right\}.
 \end{aligned}$$

Eqns. (17) and (18) also imply that:

$$(46) \quad if\mathfrak{U} = -H \left(\frac{\partial p_b}{\partial x} + i \frac{\partial p_b}{\partial y} \right) - \left(\frac{\partial \chi}{\partial y} + i \frac{\partial \chi}{\partial x} \right) + \tau^x + i\tau^y - \tau_b^x - i\tau_b^y.$$

Therefore, for an unstratified fluid ($\chi = 0, \nabla p_b = \nabla p_s$), specifying the surface pressure gradient ∇p_s closes the problem. We compute the surface stress from (46) by substituting (41). Then $u(z), v(z), U, V$ and the force balance can all be computed. See Figure 1 for an example calculation. Alternatively, the surface stress can be specified, and the pressure gradient can be computed (see `Example_barotropic_GeostrophicFlow_v0.3.ipynb`). In practice, it seems that the bottom stress always vanishes. Why?? Is this correct??

For a stratified fluid, b and hence χ (from (19)) are specified. The total pressure field $p(x, y, z)$ consists of a known buoyancy-driven piece (baroclinic) and a surface pressure driven piece from (28). Similarly, the bottom pressure field $p_b(x, y)$ consists of a known buoyancy-driven piece and a surface pressure driven piece from (29). Thus, (46) leads to:

$$\begin{aligned}
 if\mathfrak{U}(x) = & -H(x) \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] \\
 (47) \quad & - \left(\frac{\partial}{\partial y} + i \frac{\partial}{\partial x} \right) \left[- \int_{-H(x)}^0 z' b(x, z') dz' \right] + \tau_s(x) - \tau_b(x).
 \end{aligned}$$

Substituting for $\mathfrak{U}$ from (45) and rewriting $\phi^2 = f/(\nu_0 \epsilon^2)$ gives:

$$\begin{aligned}
 & \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi \xi} \left(e^{\sqrt{i}\varphi \xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) p(\xi; x) d\xi \right. \\
 & \left. - \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = -H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] \\
 (48) \quad & - (\partial_y + i\partial_x) \left[- \int_{-H(x)}^0 z' b(x, z') dz' \right] + \tau_s(x) - \tau_b(x).
 \end{aligned}$$

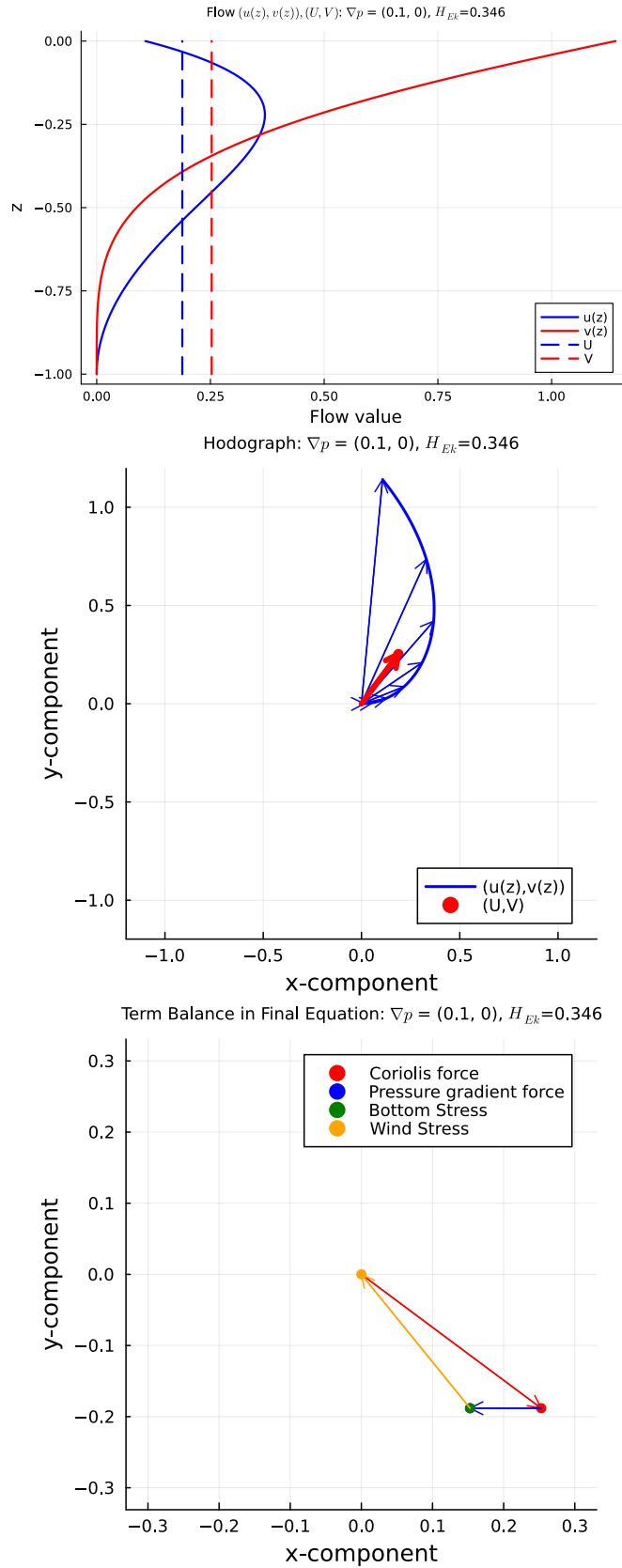


FIGURE 1. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`.

And substituting for the bottom stress τ_b from (41)

$$\begin{aligned} & \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) p(\xi; x) d\xi \right. \\ & \quad \left. - \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = \tau_s(x) - H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] \\ & \quad - (\partial_y + i\partial_x) \left[- \int_{-H(x)}^0 z' b(x, z') dz' \right] \\ & \quad - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) p(\xi; x) d\xi + \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} \tau_s(x). \end{aligned}$$

Thus, substituting for the hydrostatic pressure from (28):

$$\begin{aligned} & \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \times \\ & \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \right. \\ & \quad \left. - \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = \tau_s(x) - H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] \\ & \quad - (\partial_y + i\partial_x) \left[- \int_{-H(x)}^0 z' b(x, z') dz' \right] \\ & \quad - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \\ & \quad + \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} \tau_s(x). \end{aligned}$$

Rearranging to factor $\tau_s(x)$ terms on the right gives:

$$\begin{aligned} & \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \times \\ & \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \right\} \\ & = 2\tau_s(x) - H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] \\ & \quad - (\partial_y + i\partial_x) \left[- \int_{-H(x)}^0 z' b(x, z') dz' \right] \\ & \quad - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi. \end{aligned}$$

Hence, manipulating,

$$\begin{aligned}
& \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \\
&= \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[2\tau_s(x) - H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] + (\partial_y + i\partial_x) \int_{-H(x)}^0 z' b(x, z') dz' \right] \\
&\quad - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi.
\end{aligned}$$

Hence, breaking apart the integrals:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) d\xi \\
&+ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi \\
&= \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[2\tau_s(x) - H(x) (\partial_x + i\partial_y) \left[p_s(x) + \int_{-H(x)}^0 b(x, z') dz' \right] + (\partial_y + i\partial_x) \int_{-H(x)}^0 z' b(x, z') dz' \right] \\
&\quad - (\partial_x + i\partial_y) p_s(x) \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi \\
&\quad - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi.
\end{aligned}$$

Moving the surface pressure terms to the left gives:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi + \right. \\
&\quad \left. \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) H(x) \right\} \\
&= \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[2\tau_s(x) - H(x) (\partial_x + i\partial_y) \int_{-H(x)}^0 b(x, z') dz' + (\partial_y + i\partial_x) \int_{-H(x)}^0 z' b(x, z') dz' \right] \\
&\quad - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi \\
&\quad - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi
\end{aligned}$$

Manipulating the terms gives:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi + \right. \\
& \left. \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) H(x) \right\} \\
& = \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[2\tau_s(x) - H(x) (\partial_x + i\partial_y) \int_{-H(x)}^0 b(x, z') dz' + (\partial_y + i\partial_x) \int_{-H(x)}^0 z' b(x, z') dz' \right] - \\
& \int_{-H(x)}^0 \left[e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) \right] (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi
\end{aligned}$$

Now define $A(x, \xi)$ as

$$\begin{aligned}
A(x, \xi) & \equiv e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H} \right) \left(e^{\sqrt{i}\varphi(\xi+H)} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H)} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) \\
& = e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi(2\xi+H)} - e^{\sqrt{i}\varphi(\xi+2H)} - e^{\sqrt{i}\varphi\xi} + e^{\sqrt{i}\varphi H} \right) + e^{-\sqrt{i}\varphi(\xi-H)} + e^{\sqrt{i}\varphi(\xi+H)} \\
& = 2e^{\sqrt{i}\varphi(\xi+H)} - e^{2\sqrt{i}\varphi H} - 1 + 2e^{\sqrt{i}\varphi(H(x)-\xi)}.
\end{aligned}$$

Thus,

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \left[\int_{-H(x)}^0 A(x, \xi) d\xi + \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) H(x) \right] \\
& = \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[2\tau_s(x) - H(x) (\partial_x + i\partial_y) \int_{-H(x)}^0 b(x, z') dz' + (\partial_y + i\partial_x) \int_{-H(x)}^0 z' b(x, z') dz' \right] \\
& - \int_{-H(x)}^0 A(x, \xi) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi.
\end{aligned}$$

This needs checking! See `PlanetaryGeostrophicTheory_v0.1.ipynb`.

This final expression is of the form

$$(49) \quad (\partial_x + i\partial_y) p_s(x) = \frac{C(x)}{D(x)},$$

where $C(x)$ is given by $\tau_s(x), b(x, z), H(x)$ and $D(x)$ is given by $H(x)$. The formal solution of (49) is

$$(50) \quad p_s(x, y) = \int_0^{T(x, y)} \frac{C(x-t, y-it)}{D(x-t, y-it)} dt,$$

or, equivalently (because p_s is real)

$$(51) \quad \nabla^2 p_s(x, y) = \Re \left[(\partial_x - i\partial_y) \left(\frac{A(x, y)}{B(x, y)} \right) \right].$$

The boundary condition is that p_s is a constant on the boundary and that the global integral of p_s vanishes.

Next steps are to solve for:

- (1) $b = 0, H = 1$ in an infinite plane.
- (2) $b = 0, H(x, y) = 1 - x^2 - y^2$ in a circular parabolic basin.

- (3) $b(x, y)$ given by solving the vertical diffusive buoyancy equation and $H(x, y) = 1 - x^2 - y^2$ in a circular parabolic basin.

REFERENCES

- Klinger, B. A., and T. W. N. Haine (2019), *Ocean Circulation in Three Dimensions*, 1st ed., Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
- Nøst, O. A., and P. E. Isachsen (2003), The large-scale time-mean ocean circulation in the Nordic Seas and Arctic Ocean estimated from simplified dynamics, *J. Mar. Res.*, *61*(2), 175–210, doi:10.1357/002224003322005069.
- Peterson, H. G., and J. Callies (2023), Coupling between abyssal boundary layers and the interior ocean in the absence of along-slope variations, *J. Phys. Oceanogr.*, *53*(1), 307–322, doi:10.1175/jpo-d-22-0082.1.
- Peterson, H. G., and J. Callies (in review), Connecting the local response to abyssal mixing to the basin-scale circulation, *J. Phys. Oceanogr.*
- Salmon, R. (1998), *Lectures on Geophysical Fluid Dynamics*, Oxford University Press, doi:10.1093/oso/9780195108088.001.0001.
- Waldman, R., and H. Giordani (2023), Ocean barotropic vorticity balances: Theory and application to numerical models, *J. Advances Model. Earth Sys.*, *15*(4), doi:10.1029/2022ms003276.